ENTERPRISE TECHNICAL BLUEPRINT & SYSTEM ARCHITECTURE

THQ ERP

Full-System Technical Specification: Flutter Monorepo, PostgreSQL 17 Live Schema, Deno Edge Functions, Hardware Device Activation, Double-Entry Accounting & Resilient Offline POS

Parameter	Verified System Configuration
Repository	thariqmanchara321/THQ-ERP (Branches: main & feat/gst-nongst-final-architecture)
Production Baseline	v8.2.2 Build 19 (Windows Desktop, Android Mobile, Web Admin)
Database Engine	PostgreSQL 17.6 on Supabase (yguzrxcdvyimjrfvtcvp, Mumbai, India)
Schema Scope	269 Tables/Views, 3,482 Columns, 2,074 Constraints, 770 Indexes
Security Model	100% RLS Coverage on Public Tables, SHA-256 Device Hardware Binding
Offline Resilience	Embedded SQLite 3 WAL Mode with Local Invoicing & Idempotent FIFO Sync
Inspection Verification	Verified against live repository code, migrations, and remote database

1. Executive Summary & Core Architectural Tenets

THQ ERP is an enterprise resource planning and point-of-sale platform specifically designed for high-volume retail, wholesale distribution, manufacturing, material yards, restaurant hospitality, and logistics fleets. Built with Flutter for client applications and powered by PostgreSQL 17 on Supabase, the system provides multi-tenant isolation, double-entry financial integrity, authoritative tax compliance, and zero-downtime offline continuity.

[ARCHITECTURAL MANDATE] *Every transaction must preserve multi-tenant isolation, record auditable history, maintain balance-sheet equality, and never expose production credentials. The offline architecture guarantees that POS terminals continue billing through network disruptions, synchronizing automatically without manual intervention.*

Tenant & Store Isolation: Every operational record is bound to a tenant_id and location_id. Cross-tenant leakage is physically prevented at the database engine level via PostgreSQL Row Level Security (RLS).

Double-Entry Accounting Truth: Every commercial sale, purchase, expense, return, or payment automatically writes balanced debits and credits to the general journal (journal_entries and journal_lines). Operational screens display business records, while accounting ledgers maintain mathematical proof.

Authoritative GST Engine Separation: GST and Non-GST transaction flows operate on separate, isolated architectural pathways (v716). Tax snapshots, e-Way bills, and HSN calculations are cryptographically snapshotted upon document posting.

Offline-First Resilience: The POS terminal utilizes an embedded SQLite database in WAL (Write-Ahead Logging) mode. Transactions are queued locally, assigned deterministic UUID request IDs, and synced via a FIFO worker with automated conflict resolution.

Hardware Device Binding: Workstations and cash registers require physical activation before accepting user login. A cryptographically generated device secret is stored in local OS secure storage and verified against SHA-256 hashes in PostgreSQL.

High-Density Compact UI: Data tables and operational work queues take precedence over oversized decorative cards. Consistent keyboard navigation and high-density layouts keep cashier checkout and order entry rapid and ergonomic.

2. Monorepo & Applications Architecture

The repository is organized as a unified Dart/Flutter monorepo containing 5 dedicated client applications and 3 shared packages. This ensures atomic versioning and shared domain contracts across desktop, mobile, and web targets.

Application / Package	Size	Target	Core Responsibilities
THQ Business (client_app)	226 Dart Files	Windows x64, Web	Comprehensive ERP desktop client: Accounting, GST, Inventory, Material Yard, Purchases, Sales, Logistics, Workforce
THQ POS (pos_app)	139 Dart Files	Windows x64	High-speed retail & restaurant checkout terminal: Offline SQLite WAL engine, Cashier Shifts, KOTs, Barcode Scanning
THQ Admin Panel (admin_panel)	53 Dart Files	Web, Windows	Platform Superadmin console: Tenant Provisioning, Subscription Tiers, Device Lifecycles, App Releases, Platform Audit
THQ Client Mobile (client_mobile)	28 Dart Files	Android, iOS	Executive mobile companion: Real-time sales telemetry, stock balance checks, purchase order approvals, material load verification
THQ Mobile POS (mobile_pos)	30 Dart Files	Android, iOS	Handheld POS & waiter terminal: Table ordering, KOT printing, Bluetooth thermal printing, offline queue sync
erp_core (packages/erp_core)	19 Dart Files	Shared Foundation	Domain contracts, version contract (v8.2.2+19), CFT measurement engine, session management, updater coordinator

thq_ui (packages/thq_ui)	42 Dart Files	Shared Design System	Futuristic design tokens, virtualized high-density data tables, brand experience, dialogs, theme providers
thq_logistics (packages/thq_logistics)	4 Dart Files	Shared Logistics	Transit route contracts, vehicle run models, stop evidence specifications, driver allocation types

3. Business Hierarchy & Access Control

THQ ERP models an enterprise organization as a multi-tier hierarchy ensuring physical data isolation, granular role-based permissions, and device-level hardware governance.

Tenant / Business: The top-level legal and commercial entity (table: public.tenants). All accounting, users, products, and configurations are completely isolated by tenant_id. A tenant operates under a specific subscription plan and tax mode (GST or Non-GST).

Main Location: The authoritative headquarters or primary store (public.business_locations where is_main = true). Generated automatically upon tenant creation via v500_ensure_main_location. Holds the primary chart of accounts and default stock balances.

Child Location: Secondary commercial sites such as warehouses, material yards, transit depots, or retail branches. Child locations maintain separate stock balance ledgers (location_stock_balances) and have location-specific document sequences.

Client System (THQ Business): A Windows workstation running apps/client_app. Authorized for back-office accounting, purchasing, financial voucher creation, inventory management, and reporting. Requires device activation before employee login.

POS Terminal (THQ POS): A Windows cash register running apps/pos_app. Optimized for high-frequency barcode scanning, cash drawer kicking, and receipt printing. Operates with an independent local SQLite offline cache and dedicated device invoice series.

Device Activation: Physical machines must claim an activation token issued by the business owner. A unique device secret is stored in local secure storage, preventing unauthorized access even if credentials are compromised.

4. Live PostgreSQL 17 Database Architecture

The live database operates on PostgreSQL 17.6 on Supabase (yguzrxcdvyimjrfvtcvp) with 269 tables/views, 3,482 columns, 770 indexes, 79 RLS policies, 1,312 functions/RPCs, and 426 triggers.

Business Area	Tables	Sample Entities
Products and inventory	35	inventory_batch_balances_v483, inventory_batches_v483, inventory_serials_v483, inventory_trace_events_v483, ... (+31 more)
Logistics and transportation	32	aggregate_load_events_v617, aggregate_loads_v617, aggregate_order_events_v618, aggregate_order_lines_v618, ... (+28 more)
Platform administration	26	platform_audit_log, tracking_conversions_v633, tracking_document_lines_v633, tracking_return_bridges_v633, ... (+22 more)
Accounting and finance	25	accounting_account_mappings, accounting_accounts, aggregate_freight_settlements_v620, bank_accounts_v500, ... (+21 more)
Locations and devices	24	aggregate_direct_transit_locations_v621, business_devices, business_locations, customer_receipt_allocations, ... (+20 more)
Business and tenants	20	business_division_members, business_divisions, business_task_comments_v500, business_task_history_v500, ... (+16 more)
Purchases and suppliers	20	goods_receipt_items_v484, goods_receipts_v484, purchase_invoice_items_v484, purchase_invoices_v484, ... (+16 more)
Users and permissions	19	platform_admin_role_assignments, platform_admins, aggregate_vehicle_profiles_v617, business_user_app_access, ... (+15 more)
GST and taxation	16	gst_authoritative_tx_context_v520, gst_transaction_requests_v520, gst_document_line_snapshots_v520, gst_document_snapshots_v520, ... (+12 more)
Sales and billing	13	cash_drawer_movements, cashier_shift_edits, cashier_shifts, document_origins, ... (+9 more)
Audit and tracking	10	app_error_logs, audit_findings_v600, audit_risk_config_v600, business_audit_log, ... (+6 more)
Restaurant	9	restaurant_kot_items, restaurant_kots, restaurant_order_items, restaurant_order_modifiers, ... (+5 more)
POS and offline synchronization	6	mobile_pos_kot_requests_v520, mobile_pos_kot_requests_v488, mobile_pos_terminal_settings_v488, pos_offline_sync_v486, ... (+2 more)
Workshop and services	5	service_jobs, service_vehicles, workshop_job_cards, workshop_job_items, ... (+1 more)
Staff and attendance	5	staff_attendance_v630, staff_earnings_v630, staff_members_v630, staff_payments_v630, ... (+1 more)
Customers and CRM	4	customer_groups_v500, customer_loyalty_ledger_v500, customer_prices_v482, customers

5. The 15 Main Business Workflows

All operational events in THQ ERP trace through application screens, services, database RPCs, and tables:

Workflow 1: Business Registration and Owner Activation

App & Screens: THQ Admin Panel (admin_panel) | create_business_screen.dart, create_owner_screen.dart

Services & RPCs: manage-business-users-v31 (Edge Function), createClient | assign_business_code(), v500_ensure_main_location(p_tenant_id), v600_seed_role_permissions()

Tables Affected: tenants, profiles, tenant_memberships, roles, user_roles, business_locations, accounting_accounts

Sequence Flow: Superadmin fills business details -> Edge function creates tenant record -> assign_business_code generates unique uppercase code -> v500_ensure_main_location creates 'Main Branch' -> Chart of accounts seeded -> Owner user registered in auth.users and public.profiles -> Activation code generated.

Workflow 2: User Login and Device-Bound Authentication

App & Screens: THQ Business & THQ POS (client_app, pos_app) | client_login_screen.dart, pos_login_screen.dart

Services & RPCs: ClientAuthService, username-login (Edge Function) | system_claim_activation_v47, gotrue.signInWithPassword

Tables Affected: business_devices, system_installations, user_login_names, tenant_memberships, business_user_location_access

Sequence Flow: Cashier enters short username + password -> App reads deviceId + deviceSecret from secure storage -> username-login validates deviceSecret SHA-256 against business_devices.device_secret_hash -> Scoped lookup maps username to internal auth email -> Supabase Auth generates JWT session -> Location permissions resolved.

Workflow 3: Product Creation, Multi-Unit & Pricing Policies

App & Screens: THQ Business (client_app) | add_product_screen.dart, inventory_products_screen.dart

Services & RPCs: InventoryService, CommercialPricingService | product_identifiers_v482 trigger, v628_batch_price_override

Tables Affected: products, product_variants, product_units_v481, price_lists_v482, gst_product_tax_profiles_v520

Sequence Flow: User defines product, base unit (e.g. CFT, PCS, KG), HSN code, tax rate, and pricing tiers -> Product and variants inserted -> Secondary conversion units mapped in product_units_v481 -> Barcode/SKU sequence generated -> GST tax profile bound in gst_product_tax_profiles_v520.

Workflow 4: Purchase Order, Requisition & Goods Receipt (GRN)

App & Screens: THQ Business (client_app) | goods_receipts_v484, purchase_orders_v480 screens

Services & RPCs: InventoryService, ApprovalService | v480_purchase_order_create, v484_goods_receipt_post

Tables Affected: purchase_orders_v480, purchase_order_items_v480, goods_receipts_v484, goods_receipt_items_v484, stock_movements

Sequence Flow: Requisition approved -> PO generated and sent to supplier -> Physical goods arrive at location -> GRN recorded with received quantities, batches, and serials -> Stock balance updated in location_stock_balances -> PO status updated to 'received' or 'partially_received'.

Workflow 5: Purchase Invoice & Supplier Ledger Payment

App & Screens: THQ Business (client_app) | purchase_invoices_v484, supplier_payments_v484 screens

Services & RPCs: AccountingService, ExpenseService | v484_purchase_invoice_post, v484_supplier_payment_allocate

Tables Affected: purchase_invoices_v484, purchase_invoice_items_v484, suppliers, supplier_payments_v484, supplier_ledger_entries_v484, journal_entries, journal_lines

Sequence Flow: Supplier bill entered matching GRN -> Tax credits calculated -> Accounting journal posted: Debit Inventory/Purchase Expense, Debit GST Input Tax Credit (CGST/SGST/IGST), Credit Accounts Payable -> Supplier payment disbursed (Bank/Cash) -> Ledger balance cleared.

Workflow 6: Sales & POS Checkout Billing

App & Screens: THQ POS & THQ Business (pos_app, client_app) | pos_screen.dart, client_entry_screen.dart

Services & RPCs: PosCompletionService, OfflinePosService | pos_complete_invoice_v522, assign_device_document_number

Tables Affected: sales, sale_items, location_stock_balances, stock_movements, gst_document_snapshots_v520, journal_entries, journal_lines

Sequence Flow: Cashier scans items -> Price resolved -> Payment tendered -> Unique invoice number assigned via document counter -> Stock decremented in location_stock_balances -> GST document snapshot stored -> Double-entry journal posted: Debit Cash/Bank/Receivable, Credit Sales Revenue, Credit GST Output Tax Liability.

Workflow 7: Split Payments & Multi-Tender Allocation

App & Screens: THQ POS (pos_app) | payment_center_screen.dart, pos_screen.dart

Services & RPCs: PosCompletionService | sale_payment_allocations_v522

Tables Affected: sale_payments, sale_payment_allocations_v522, cash_drawer_movements, customer_receipts

Sequence Flow: Customer splits bill (e.g. 50% Cash, 50% UPI) -> Distinct payment lines created in sale_payments -> Cash drawer kicks and logs movement -> Journal lines debit respective Cash Account and Bank/UPI Clearing Account independently.

Workflow 8: Inventory Movement, Stock Balancing & Traceability

App & Screens: THQ Business (client_app) | inventory_movement_history_screen.dart, aggregate_yard_stock_screen.dart

Services & RPCs: InventoryService | v483_inventory_trace_capture, stock_movements trigger

Tables Affected: stock_movements, location_stock_balances, inventory_batches_v483, inventory_serials_v483, inventory_trace_events_v483

Sequence Flow: Stock movement occurs (Sale, GRN, Transfer, Adjustment) -> Immutable row written to stock_movements -> Database trigger recalculates location_stock_balances atomically -> Batch expiry and serial lifecycle updated -> Audit trace event emitted.

Workflow 9: Sales Returns & Purchase Returns

App & Screens: THQ Business & POS (client_app, pos_app) | sales_returns, purchase_returns screens

Services & RPCs: AccountingService, PosCompletionService | v4_sales_return_accounting_trigger, v4_reverse_source_journal

Tables Affected: sales_returns, sales_return_items, purchase_returns, purchase_return_items, location_stock_balances, journal_entries, journal_lines

Sequence Flow: Customer returns item -> Sales return bill issued referencing original invoice -> Returned goods restored to stock balance -> Reverse accounting entry posted: Debit Sales Returns, Debit GST Output Tax (reversal), Credit Cash/Accounts Receivable.

Workflow 10: Double-Entry Accounting & Automatic Journal Generation

App & Screens: THQ Business (client_app) | accounting_screen.dart, daily_accounts_report_view.dart

Services & RPCs: AccountingService, AccountingWorkspaceService | accounting_commit_guard_v600, v500_assert_posting_period_open

Tables Affected: accounting_accounts, journal_entries, journal_lines, financial_years_v500

Sequence Flow: Operational transaction posts -> Trigger checks posting period is open -> Balanced journal_entries header created -> Multiple journal_lines written ensuring sum(debit) == sum(credit) -> Real-time general ledger, trial balance, and P&L; updated instantly.

Workflow 11: Authoritative GST Invoicing, Series & Filing

App & Screens: THQ Business (client_app) | gst_authoritative_screen.dart, gst_portal_import_screen.dart

Services & RPCs: GstV520RouteGuard | gst_authoritative_tx_context_v520, gst_invoice_series_v632

Tables Affected: gst_registrations_v520, gst_document_snapshots_v520, gst_document_line_snapshots_v520, gst_return_periods_v600, gst_portal_itc_documents_v600

Sequence Flow: Invoice evaluates supply type (Intra-state: CGST+SGST, Inter-state: IGST) -> Document number drawn from dedicated GST series -> Immutable snapshot stored with exact taxable value and cess -> Portal ITC documents ingested for GSTR-2B reconciliation.

Workflow 12: Offline POS Billing & Deterministic Synchronization

App & Screens: THQ POS (pos_app) | offline_pos_sync_screen.dart, pos_screen.dart

Services & RPCs: OfflinePosService, OfflinePosSyncService | pos_offline_product_cache_v486, pos_complete_invoice_v522, v520_offline_sync_guard

Tables Affected: offline_invoices (SQLite), pos_offline_sync_v486, sales, transaction_requests_v47

Sequence Flow: Internet drops -> OfflinePosService continues billing using local SQLite cache -> Invoices assigned local sequence and UUID request_id -> When network returns, OfflinePosSyncService flushes queue -> Remote sync guard guarantees exactly-once processing.

Workflow 13: Hardware Device Registration & Activation

App & Screens: THQ POS & THQ Business (pos_app, client_app) | device_activation_screen.dart

Services & RPCs: device-activate (Edge Function) | system_claim_activation_v47

Tables Affected: business_devices, system_installations

Sequence Flow: Admin generates activation code -> Physical workstation enters business code + activation code -> Edge function computes SHA-256 hash -> system_claim_activation_v47 creates business_devices row -> App receives raw device_secret and persists to OS secure storage.

Workflow 14: Audited Business Deletion & Cascade Lifecycle

App & Screens: THQ Admin Panel (admin_panel) | business_details_screen.dart

Services & RPCs: delete-business-v41 (Edge Function) | delete-business-v41, business_audit_write

Tables Affected: tenants, tenant_memberships, business_devices, private.platform_audit_log

Sequence Flow: Superadmin requests tenant deletion with confirmation token -> Edge function verifies platform admin role -> Tenant status set to 'deleting' -> Child devices and user bindings revoked -> Full audit trail recorded in private.platform_audit_log.

Workflow 15: Automated Application Updates & Inno Setup Deployment

App & Screens: THQ Business & THQ POS (client_app, pos_app) | support_screen.dart, updates_dialog.dart

Services & RPCs: SoftwareUpdateVerifier, app-release-download (Edge Function) | app_release_check_v820, platform_app_update_logs insert

Tables Affected: platform_app_releases, platform_app_update_logs

Sequence Flow: App checks app_release_check_v820 -> New release found -> Signed installer downloaded from private storage -> SHA-256 and Authenticode signature validated fail-closed -> thq_update_coordinator.ps1 launches detached -> Inno Setup runs with UAC elevation -> App restarts.

6. Supabase Edge Functions Architecture

Analysis of deployed Deno Edge Functions on Supabase versus repository source control:

Slug	JWT	Inputs	Responsibility & Tables
device-activate	False (Public)	business_code, activation_code, installation_id, app_key	Issues device secrets via SHA-256 tokens; claims hardware activation.
username-login	False (Public)	username, password, app_key, device_id, device_secret	Translates human usernames to auth emails after validating device hardware hashes.
app-release-download	False (Signed)	app_key, version, platform, build_number, token	Proxies secure installer downloads from private storage buckets.
manage-business-users-v31	True (Auth)	action, tenant_id, user_id, role_ids, location_ids	Superadmin & tenant owner user management and role assignment.
manage-tenant-users-v32	True (Auth)	tenant_id, target_user, division_id	Division-scoped tenant user assignment.
delete-business-v41	True (Admin)	tenant_id, confirm_name, reason	Guarded, audited deletion of tenant organizations.
thq-api	True (Auth)	REST v1 resources: /orders, /inventory, /webhooks	External integration API gateway for third-party systems.

7. Offline Storage & Synchronization Architecture

THQ POS utilizes SQLite 3 (thq_pos_offline_v486.sqlite) in WAL mode to ensure continuous, fail-safe billing:

WAL Mode Persistence: PRAGMA journal_mode=WAL and PRAGMA synchronous=FULL guarantee zero transaction loss even during abrupt hardware power failure.

Authoritative Local Caching: The local catalog cache stores products, variants, customer balances, serials, and tax rates via pos_offline_product_cache_v486.

Deterministic Idempotency: Every invoice is assigned an RFC-4122 v4 UUID request_id and payload hash. The server RPC checks transaction_requests_v47 to prevent duplicate posting.

Conflict Resolution: Discrepancies in stock or prices trigger status = 'conflict' and notify the cashier via offline_pos_sync_screen.dart, preventing silent data corruption.

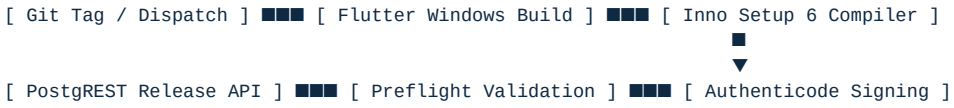
8. Security & Technical Architecture Audit

Categorized audit findings across live backend, RLS, and functions:

Classification	Finding	Analysis & Remediation
Verified Issue	4 Legacy Edge Functions Active on Remote Backend	create-business-owner, manage-business-users, hyper-endpoint, and gst-provider-worker remain deployed on Supabase but are superseded in Git. Remediation: Decommission via supabase functions delete.
Potential Risk	Commercial Code Signing Requirement	In-app updates enforce fail-closed Authenticode validation. Installers signed with the internal QA test certificate will fail on customer PCs lacking the QA root. Remediation: Supply commercial OV/EV PFX in GitHub secrets.
Potential Risk	Public Telemetry Insert Rate Limiting	platform_app_update_logs permits anonymous inserts so updaters can report fatal launch failures. Remediation: Add PostgreSQL IP rate limiting to prevent telemetry flooding.
Verified Clean	100% RLS Coverage on Public Tables	All 260 public tables enforce rowsecurity = true. Tables without direct client policies default to fail-closed deny-all access, accessible only via security definer RPCs.
Verified Clean	Hardware Device Binding Enforcement	Physical workstations require activation tokens. Passwords alone cannot authenticate on unapproved hardware.

9. DevOps, Build Automation & CI/CD

Automated packaging and release pipelines defined in .github/workflows/thq-release-windows.yml:



10. Comprehensive Technical Glossary

CFT (Cubic Feet): Standard volumetric measurement for sand, aggregates, and raw materials. Calculated from length, width, and height with heap adjustments.

GRN (Goods Receipt Note): Authoritative warehouse intake document verifying delivered physical quantities against purchase orders prior to invoice posting.

HSN / SAC: Harmonized System of Nomenclature code classifying goods and services for authoritative GST rate determination in India.

KOT (Kitchen Order Ticket): Restaurant order dispatch sent to kitchen preparation stations before commercial billing.

RLS (Row Level Security): PostgreSQL security feature restricting row visibility based on the executing user's authenticated role and tenant context.

WAL (Write-Ahead Logging): SQLite journaling mode offering superior concurrency and crash-safe transaction resilience on POS terminals.

UAC (User Account Control): Windows security infrastructure ensuring installers elevate privileges safely to Program Files directories.